文献复现进度汇报

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文献

Tian Hongda,Li Wanqing,Ogunbona Philip O,Wang Lei. **Detection and Separation of Smoke From Single Image Frames**.[J]. IEEE transactions on image processing: a publication of the IEEE Signal Processing Society,2018,27(3).

学习烟雾字典

- 648张烟雾图片100*100*3批量裁剪，变灰度，标准化，
- 学习烟雾字典时，稀疏编码方式先使用omp，后面再用sklearn.lasso方法对比

```
C:\Users\28070\AppData\Roaming\Python\Python37\site-packages\sklearn\linear_model\_omp.py:387: RuntimeWarning:  Orthogonal matching pursuit ended prematurely due to linear
dependence in the dictionary. The requested precision might not have been met.

  copy_X=copy_X, return_path=return_path)
```

```
self._initialize(y)
for i in range(self.max_iter):
    print('第%d次迭代'%i)

    x = linear_model.orthogonal_mp(self.dictionary, y, n_nonzero_coefs=self.n_nonzero_coefs)
    e = np.linalg.norm(y - np.dot(self.dictionary, x))
    print('err',e)
    if e < self.tol:
        break
    self._update_dict(y, self.dictionary, x)

self.sparsecode = linear_model.orthogonal_mp(self.dictionary, y, n_nonzero_coefs=self.n_nonzero_coefs)
return self.dictionary, self.sparsecode
```

学习烟雾字典

问题:

I'm using sklearns OrthogonalMatchingPursuit to get a sparse coding of a signal using a dictionary learned by a KSVD algorithm. However, during the fit I get the following RuntimeWarning:

```
/usr/local/lib/python2.7/dist-packages/sklearn/linear_model/omp.py:391: RuntimeWarning:  Orthogonal matching pursuit ended prematurely due to linear d
ependence in the dictionary. The requested precision might not have been met.    copy_X=copy_X, return_path=return_path)
```

In those cases the results are indeed not satisfactory. I don't get the point of this warning as it is common in sparse coding to have an overcomplete dictionary a n thus also linear dependency within it. That should not be an issue for OMP. In fact, the warning is also raised if the dictionary is a square matrix.

Might this Warning also point to other issues in the application?

回答1:

The problem was in the data vector `y` in

```
omp = OrthogonalMatchingPursuit(n_nonzero_coefs=target_sparsity) omp.fit(D, y)
```

It contained numbers with very small magnitude. When I normalize `y` as well as `D` the fit works with the expected accuracy.

训练SVM

- 648/12960张纯烟雾图片在纯烟字典得到编码，50000张非烟图片在非烟字典得到编码，导入SVM进行训练和测试
- 实验：
  - 训练：450+/35000-；测试：198+/15000-

总共错了137个数据
错误率为0.901434%
precision为100.000000%
recall为30.808081%
f1_score为47.104247%

  - 训练：9072+/35000-；测试：3888+/15000-

总共错了525个数据
错误率为2.779543%
precision为100.000000%
recall为86.496914%
f1_score为92.759619%

总共错了1019个数据
错误率为5.394960%
precision为100.000000%
recall为73.791152%
f1_score为84.919343%

SVM对单张图片进行分类

- 用前面训练好的SVM，对于含有烟雾的一张图片，滑动窗口得到图像块，对每个图像块进行检测，将图像块在纯烟字典和非烟字典得到的编码**直接相加**作为图像块的编码，导入SVM得到检测结果。
- 编码直接相加的合理性：对于一张含有烟雾的图片，得到的图像块由烟雾和非烟雾组成，理想情况非烟雾成分在烟雾字典的编码为0，因此如果图像块在烟雾字典的编码不为0，说明图像中含有烟雾成分。

SVM分类结果

0 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0
0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 1 0 1 1 1 0 0 0 0 0 1 0 0 0 0
0 0 0 0 0 0 1 0 1 0 0 0 0 0 1 0 1 0 1 0
0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0
0 1 0 0 1 0 0 0 1 0 1 0 0 0 0 0 0 0 1 1
1 0 0 0 1 0 0 0 0 1 0 0 0 0 1 0 1 0 0 0
1 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0
0 1 0 0 0 1 1 1 0 0 0 0 1 1 0 0 0 0 0 0
0 0 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0
1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

分析问题：

- SVM训练的时候，也应该对双字典得到稀疏编码，然后将得到的编码相加导入SVM

Thank You!